ACE 182 – Machines Theory

Core Course: 3 Credits (2 Lecture + 1 Tutorial)

Prerequisite(s):

Semester: Spring 2017

Instructor: Dr. Ahmed Khalifa

E-mail: ahmed.khalifa0687@gmail.com

Schedule: Thursday 10:40 - 12:45

Location: **FPGA Room**

Office: Ground floor – Control Building

Office Hrs.: 2 hrs. after lecture

Text Book: R. L. Norton, Design of Machinery "An Introduction to the Synthesis and Analysis of Mechanisms and Machines", McGraw Hill Higher Education; 3rd edition

Important Requirements

All students are required to

- Projects, and all other assignments will be submitted in electronic format.
- No homework need to be submitted, they are just for personal practice and solutions will be provided.

Assessment Tools

T1: Examinations/Tests

T3: Group Projects

T6: Student Survey

Course Objectives

The course provides students with instruction in the fundamentals of theory of machines. The theory of Machines and Mechanisms provides the foundation for the study of displacements, velocities, accelerations, and static and dynamic forces, and balancing required for (1) the proper design of mechanical linkages, (2) analysis and modification of existing mechanisms.

Course Intended Learning Outcomes

Students combine theory, graphical and analytical skills to understand the Engineering Design. Upon successful completion of the course, the student will be able:

ILO#	Description	Assessment Tool
1	Identify mechanisms and predict their motion	T1,T6
2	Calculate the degrees of freedom of mechanisms	T1,T6
3	Determine the positions, velocities and accelerations of links and points on mechanisms	T1,T6
4	Calculate dynamic joint forces of mechanisms	T1,T6
5	Balance simple rotating objects	T1,T6
6	Analyze and modify existing mechanisms	T1,T6
7	Synthesize, analyze, and simulate, a mechanism in a computer aided software	T1,T6
8	To increase the ability of students to work cooperatively on teams	T3
9	Prepare and present technical reports	T3

Topics and Schedule

Week #	Topic	Practice and Assessment
1	Introduction to the course • Rule of theory of machines • What are we doing here? • Syllabus and Course Outline • What is expected of you in this course? • Introductory Topics	
2	Kinematics Fundamentals • Mobility • Types of Motion • Links, Joints, and Kinematic Chains • Determining Degree of Freedom or Mobility	HW #02 Software Tutorial
3	Kinematics Fundamentals • Number Synthesis • Isomers • Inversions • Grashof Condition • Examples	HW #03 Software Tutorial
4	Graphical Linkage Synthesis • Definitions and Synthesis Types • Limiting Conditions	HW #04 Software Tutorial

	• Two-Position Synthesis • Three-Position Synthesis	
5	Graphical Linkage Synthesis • Quick Return Motion Mechanisms • Coupler Curves • Cognates • Straight-Line Linkages • Dwell Mechanisms • Four-bar Crank-Rocker Quick-Return Linkage for Specified Time Ratio	HW #05 Software Tutorial
6	Position Analysis • Coordinate systems • Position and Displacement • Translation, Rotation, and Complex Motion	HW #06 Software Tutorial
7	**Midterm #1**	
8	Position Analysis • Algebraic Position Analysis of linkages • The Four-bar Slider-Crank Position Solution • Position of any Point on a linkage	HW #08 Software Tutorial **Follow up #01 due**
9	Velocity Analysis • Graphical Velocity Analysis • Velocity Analysis with Instant Centers • Analytical Solutions for Velocity Analysis • Velocity of any Point on a Linkage	HW #09 Software Tutorial
10	Acceleration Analysis • Graphical Acceleration Analysis • Analytical Solutions for Acceleration Analysis • Acceleration of any Point on a Linkage	HW #10 Software Tutorial
11	Dynamics Fundamentals • Newton's Laws of Motion • Dynamic Models • Basic definitions	HW #11 Software Tutorial
12	**Midterm #02**	

13	Dynamic Force Analysis • Newtonian Solution Method • Single link in pure rotation • Force Analysis of a Four-bar linkage	**Follow up #02 due**
14	Balancing • Static Balance • Dynamic Balance • Balancing Linkages	HW #11 Software Tutorial HW #10 Software Tutorial
15	**Project Evaluation**	
16	**Final Exam**	

Assessment (Grading)

40% Final (you must score at least 50% in final to pass)
20% Midterms (2 midterms, all counted)
10% Pop quizzes (5 to 10 randomly distributed – all counted)
30% Project (Incl. 2 x 5% follow up reports)

Course Project

Topics

- The topics should include the synthesis, analysis, simulation of a mechanism.

Teams

- To register in a project, the team must submit a file with the following items to your course instructor's email address before the 5th weak:
  - Names of team members, IDs, email address.
  - Project title
  - Project introduction (motivation and objective)

Project Delivery: 3 days before the project presentation due

- Before the presentation and to complete evaluating the project, the team must prepare a CD/Memory stick for the course instructor containing all materials related to the project and a well-documented project technical report.

Project Presentation: 15th weak

Project Evaluation:

- The project team presents the developed project in 30 minutes plus 5 minutes for a question

period.

- Each presentation is attended by the course instructor, some critiquers (maybe), and the project team.
- During the presentation, the team members can share the speaking equally.
- All team members must be on the stage. Absent members need to provide documentation excusing their participation, e.g., a doctor's note.
- The critiquers will mark the presentation.

- The project evaluation will be based on the following criteria:
  - Project management (15%)
    - Planning
    - Follow up
    - Corrective actions and re-planning
  - Software (20%)
    - Open source or your own
    - Accuracy
    - Output graphics
  - Final Report (30%)
    - Technical report format (Cover, TOC, Introduction and literature survey, mathematical background, results, conclusions, reference, appendices)
    - Rigor of literature survey
    - Details of the model analysis
    - Numerical results
  - Final presentation (15%)
    - Public decimation of knowledge (Website. Wiki, videos, public slides, public reports, design graphics and charts, etc...)
    - Presentation graphics and appeal
    - Accuracy of presentation content
    - Comprehensibility of presentation content
    - Audience capturing
    - Audience participation and reply to questions
    - Timing

Course Policies

Late Submission Policy:

- Anything submitted after 11:59PM on the due date will be penalized by 50% for each 24 hours of lateness.

Absence Policy:

- Attendance in all academic activities is mandatory.

- Students who are absent more than 25% of the time in the course should not be permitted to attend the final examination.
- No makeup exams should be offered for missed exams.

Academic Integrity Policy:

Students are required to refrain from all forms of dishonorable or unethical conduct related to their academic work including;

- **Plagiarism**:

Submitting material that is in part or whole is not entirely one's own work without properly citing sources. Plagiarism includes, but is not limited to:

1. Submitting a copied piece of writing as original work
2. The quotation or other use of another person's words, ideas, opinions, thoughts, or theories (even if paraphrased into one's own words) without acknowledgment of the source
3. The quotation or other use of facts, statistics, or other data or materials (including images) that are not clearly common knowledge without acknowledgment of the source.

- **Fabrication**:

Falsifying or inventing any information, data, or citation; presenting data that were not gathered in accordance with standard guidelines defining the appropriate methods for collecting or generating data and failing to include an accurate account of the method by which the data were gathered or collected including the incorrect documentation of a source;

1. The citation, in a bibliography or other list of references, of sources that were not used to prepare the academic work.
2. The inclusion in an academic work of falsified, invented, or fictitious data or information, or the deliberate and knowing concealment or distortion of the true nature, origin, or function of such data or information.
3. The unauthorized submission of an academic work prepared totally or in part by another.

- **Cheating:**

Cheating is defined as fraud, deceit, or dishonesty in an academic assignment, or using or attempting to use materials, or assisting others in using materials that are prohibited or inappropriate in the context of the academic assignment in question.